What Slows You Down: Real-Time GPS Classification of Delay Sources on Kathmandu’s Urban Corridors

---

Problem Statement

Travel delays on Kathmandu’s roads are severe. Buses crawl at walking speed. A trip that should take twelve minutes routinely takes twenty-five. But we know almost nothing about **why a vehicle stops at any specific place and time.**

Is it a red light? A bus blocking the lane to pick up passengers? Pedestrians crossing where no zebra exists? A double-parked car forcing everyone into one lane? Street vendors spilling onto the road? A pothole everyone swerves around?

Right now, nobody in Kathmandu can answer that question systematically. The city has no network of traffic sensors. No automated monitoring at intersections. Interventions are consequently generic — widen the road, add a signal, enforce parking — without any evidence that the chosen fix addresses what is actually causing vehicles to stop on that corridor at that hour.

This research fills that gap. It classifies the real-time causes of individual stop events across three major urban corridors, at different times of day, using nothing but a smartphone GPS trace and structured observation from a moving ride-share vehicle. The output is a map not of “congestion” — but of **what, specifically, is slowing vehicles down at each bottleneck.**

Research Questions

Primary:

RQ1: What is the composition of travel delay by cause on three selected urban corridors in Kathmandu, and how does this composition vary by corridor and time of day?

Secondary:

RQ2: Which individual bottleneck locations cause the highest total delay — measured as frequency multiplied by duration — on each corridor, and what is their dominant cause?

RQ3: What proportion of total journey time is spent in a stopped state, and what proportion of that stopped time is attributable to each cause category?

RQ4: Are specific delay causes spatially clustered? For example, does bus interference concentrate near designated stops? Does market friction cluster around commercial nodes? Does pedestrian crossing delay peak near hospitals, schools, and temples?

RQ5: Can a voice-annotated smartphone GPS observation protocol — executed from a moving ride-share vehicle by a single observer — produce a classified delay-source map reliable enough to guide corridor-level intervention decisions?